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MAE 3270 HW 10 Q6 Dowling Problem 7.24

Using Tresca Theory (max shear stress theory) Prof Alan Zehnder Michelle Zhou Nov 3, 2021

```
T = 1.5*10^3; % torque in Nm
M = 1*10^3; % bending moment in Nm
```

Part a: AISI 1020 Steel

as rolled

```
fprintf('Part a \n');

sigma_o = 260*10^6; % strength in Pa for AISI 1020 steel from Table
 3.8
tau_o = sigma_o/2; % shear strength in Pa

d = [0:0.001:0.075]; % diameter list in m

c = d/2; % radius list in m
J = (pi/2)*c.^4; % polar moment of inertia list in m^4
tau_xy = T*c./J; % shear stress due to torque list in Pa

I = J/2; % moment of inertia list in m^4
sigma_x = M*c./I; % normal stress in x-dir due to bending moment in Pa
tau_max = sqrt((sigma_x/2).^2 + tau_xy.^2); % max shear stress from
 Mohr's circle in Pa

X_o = tau_o./tau_max; % dimesnionless safety factor against yield

% plotting safety factor vs thickness
close all
plot(d*10^3, X_o)
xlabel('Diameter (mm) ')
ylabel('Safety Factor Against Yielding')
title('Safety Factor Against Yielding vs Diameter (mm) for AISI 1020')

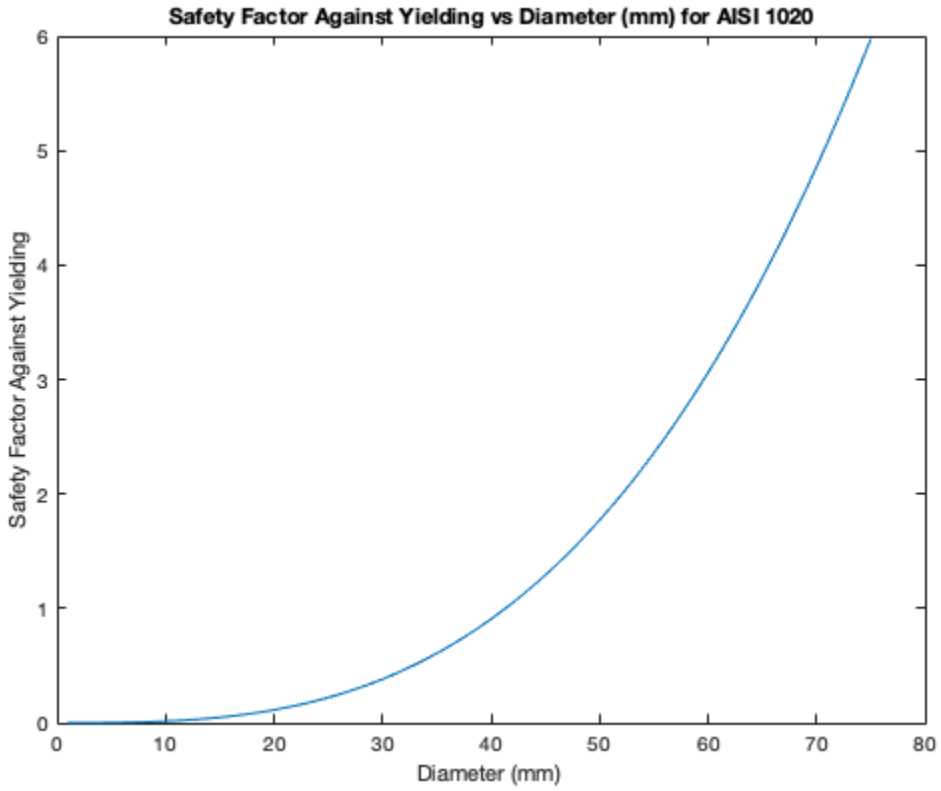
% calculate mass after picking off value from ^ graph
d_pick = 52*10^-3; % picked diameter in m
fprintf('The diameter of the solid AISI 1020 shaft would be %2.0f mm
 \n', d_pick*10^3);
```

```
r = d_pick/2; % radius in m
L = 1; % length in m
V = (pi*r^2)*L; % volume of a cylinder

rho = 7900; % density of AISI 1020 in kg/m^3 from Table 3.8
mass_AISI_1020 = rho*V;

fprintf('The mass of the solid AISI 1020 shaft would be %2.1f kg \n',
 mass_AISI_1020);
fprintf('\n');
```

Part a
The diameter of the solid AISI 1020 shaft would be 52 mm
The mass of the solid AISI 1020 shaft would be 16.8 kg



Part b 2024-T4 Aluminum

```
fprintf('\n');
fprintf('Part b \n');

sigma_o = 303*10^6; % 0.2% yield strength in tension in Pa for 2024-T4
 Aluminum from Table 3.2
tau_o = sigma_o/2; % shear strength in Pa
```

```
d = [0:0.001:0.075]; % diameter list in m

c = d/2; % radius list in m
J = (pi/2)*c.^4; % polar moment of inertia list in m^4
tau_xy = T*c./J; % shear stress due to torque list in Pa

I = J/2; % moment of inertia list in m^4
sigma_x = M*c./I; % normal stress in x-dir due to bending moment in Pa
tau_max = sqrt((sigma_x/2).^2 + tau_xy.^2); % max shear stress from
 Mohr's circle in Pa

X_o = tau_o./tau_max; % dimesnionless safety factor against yield

% plotting safety factor vs thickness
figure
plot(d*10^3, X_o)
xlabel('Diameter (mm) ')
ylabel('Safety Factor Against Yielding')
title('Safety Factor Against Yielding vs Diameter (mm) for 2024-T4
 Aluminum')

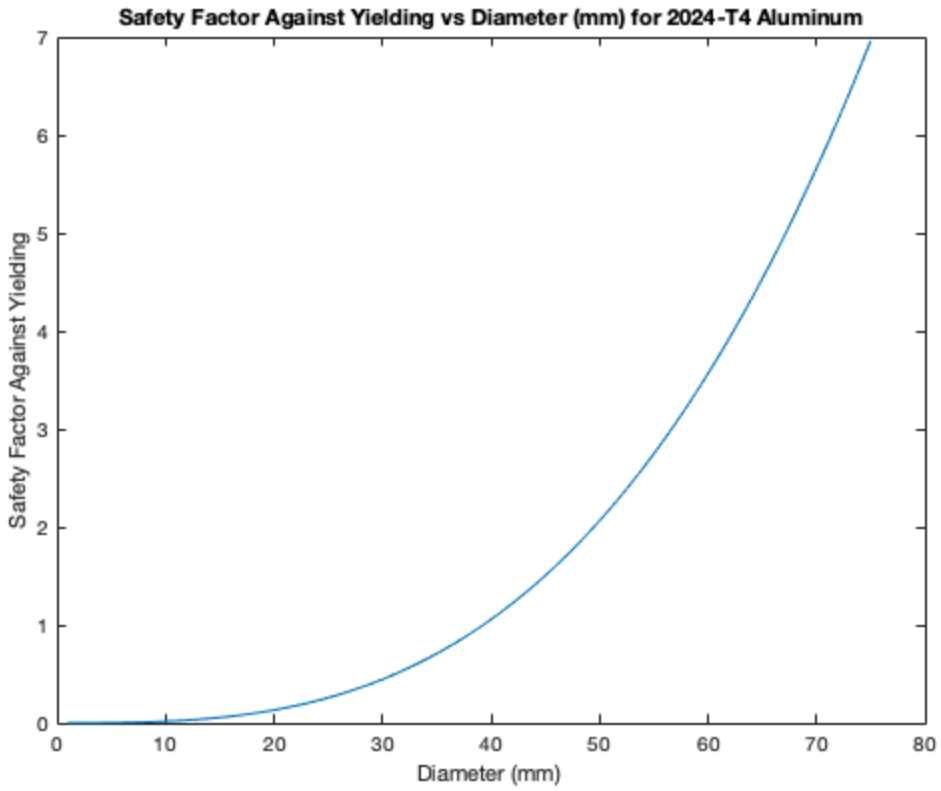
% calculate mass after picking off value from ^ graph
d_pick = 50*10^-3; % picked diameter in m
fprintf('The diameter of the solid 2024-T4 Aluminum shaft would be
 %2.0f mm \n', d_pick*10^3);

r = d_pick/2; % radius in m
L = 1; % length in m
V = (pi*r^2)*L; % volume of a cylinder

rho = 2768; % density of aluminum from Granta
mass_2024_T4 = rho*V;

fprintf('The mass of the solid 2024-T4 Aluminum shaft would be %2.1f
 kg \n', mass_2024_T4);
fprintf('\n');
```

Part b
The diameter of the solid 2024-T4 Aluminum shaft would be 50 mm
The mass of the solid 2024-T4 Aluminum shaft would be 5.4 kg



Part b: 7075-T6 Aluminum

```
sigma_o = 469*10^6; % strength in Pa for 7075-T6 Aluminum from Table
 3.8
tau_o = sigma_o/2; % shear strength in Pa

d = [0:0.001:0.075]; % diameter list in m

c = d/2; % radius list in m
J = (pi/2)*c.^4; % polar moment of inertia list in m^4
tau_xy = T*c./J; % shear stress due to torque list in Pa

I = J/2; % moment of inertia list in m^4
sigma_x = M*c./I; % normal stress in x-dir due to bending moment in Pa
tau_max = sqrt((sigma_x/2).^2 + tau_xy.^2); % max shear stress from
 Mohr's circle in Pa

X_o = tau_o./tau_max; % dimesnionless safety factor against yield

% plotting safety factor vs thickness
figure
plot(d*10^3, X_o)
xlabel('Diameter (mm) ')
ylabel('Safety Factor Against Yielding')
```

```
title('Safety Factor Against Yielding vs Diameter (mm) for 7075-T6
 Aluminum')

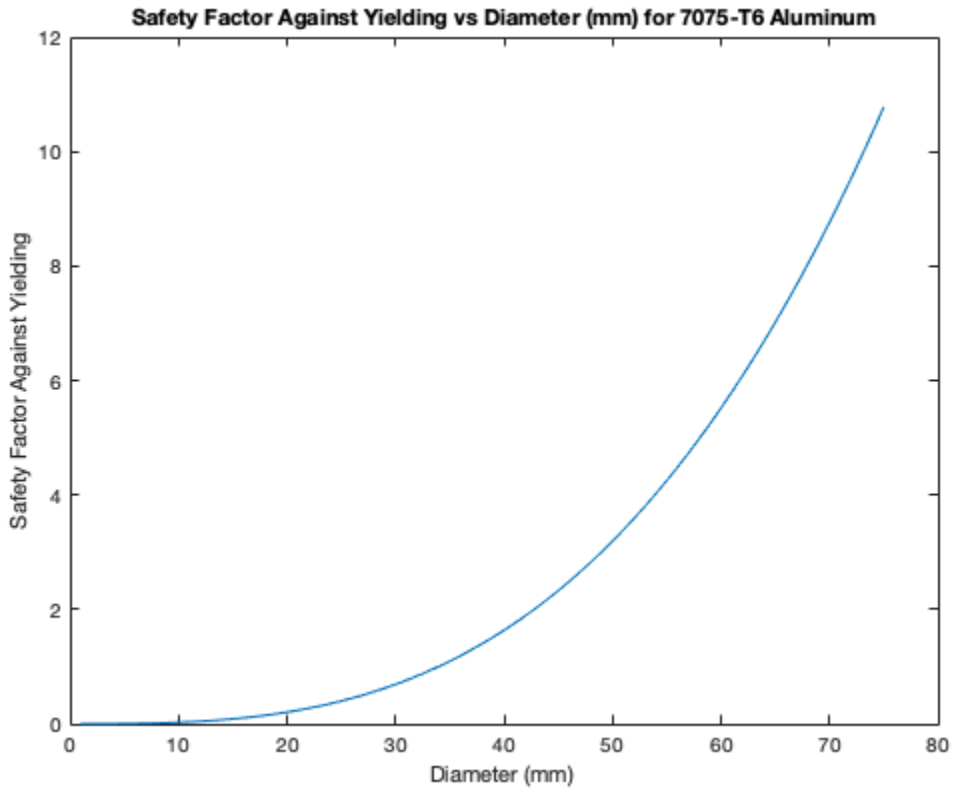
% calculate mass after picking off value from ^ graph
d_pick = 43*10^-3; % picked diameter in m
fprintf('The diameter of the solid 7075-T6 Aluminum shaft would be
 %2.0f mm \n', d_pick*10^3);

r = d_pick/2; % radius in m
L = 1; % length in m
V = (pi*r^2)*L; % volume of a cylinder

rho = 2700; % density of aluminum from Granta
mass_7075_T6 = rho*V;

fprintf('The mass of the solid 7075-T6 Aluminum shaft would be %2.1f
 kg \n', mass_7075_T6);
fprintf('\n');
```

The diameter of the solid 7075-T6 Aluminum shaft would be 43 mm
The mass of the solid 7075-T6 Aluminum shaft would be 3.9 kg



Part b: AISI 4140 Steel from Problem 4.7

I chose oil quenched and tempered at 205 deg Celsius

```
sigma_o = 1583*10^6; % strength in Pa for 140 Steel from Problem 4.7
tau_o = sigma_o/2; % shear strength in Pa

d = [0:0.001:0.04]; % diameter list in m

c = d/2; % radius list in m
J = (pi/2)*c.^4; % polar moment of inertia list in m^4
tau_xy = T*c./J; % shear stress due to torque list in Pa

I = J/2; % moment of inertia list in m^4
sigma_x = M*c./I; % normal stress in x-dir due to bending moment in Pa
tau_max = sqrt((sigma_x/2).^2 + tau_xy.^2); % max shear stress from
 Mohr's circle in Pa

X_o = tau_o./tau_max; % dimensionless safety factor against yield

% plotting safety factor vs thickness
figure
plot(d*10^3, X_o)
xlabel('Diameter (mm) ')
ylabel('Safety Factor Against Yielding')
title('Safety Factor Against Yielding vs Diameter (mm) for AISI 4140
 Steel')

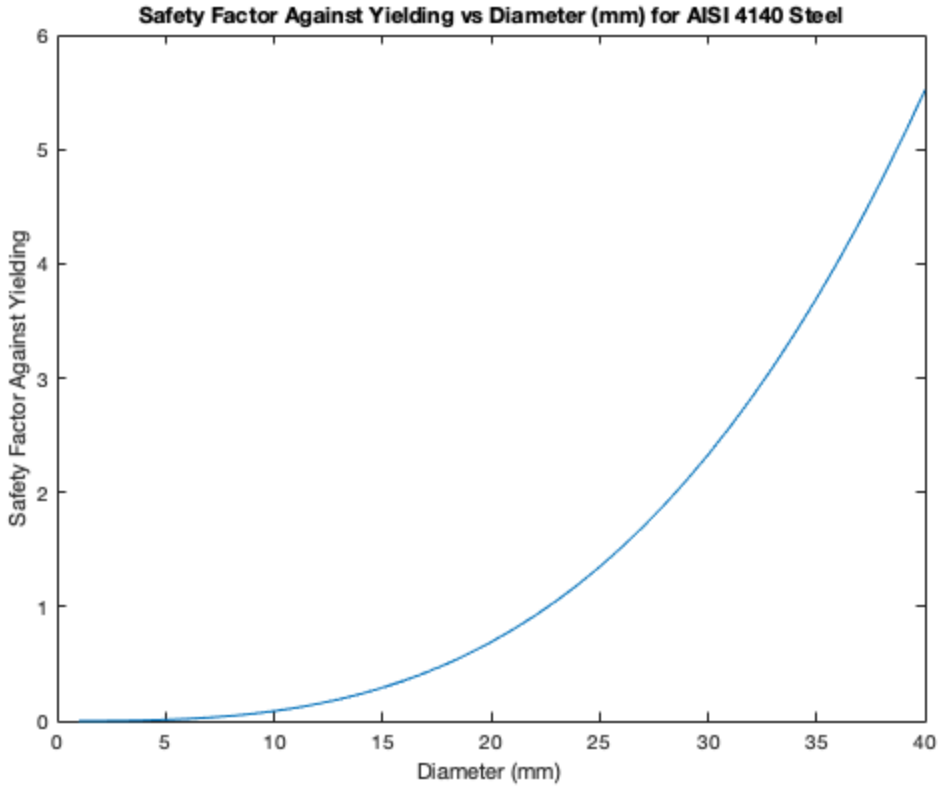
% calculate mass after picking off value from ^ graph
d_pick = 28*10^-3; % picked diameter in m
fprintf('The diameter of the solid AISI 4140 Steel shaft would be
 %2.0f mm \n', d_pick*10^3);

r = d_pick/2; % radius in m
L = 1; % length in m
V = (pi*r^2)*L; % volume of a cylinder

rho = 7889; % density of Low ally steel, AISI 4140, oil quenched and
 tempered at 205 deg C from Granta
mass_AISI_4140 = rho*V;

fprintf('The mass of the solid AISI 4140 Steel shaft would be %2.1f kg
 \n', mass_AISI_4140);
fprintf('\n');
fprintf('\n');
```

The diameter of the solid AISI 4140 Steel shaft would be 28 mm
The mass of the solid AISI 4140 Steel shaft would be 4.9 kg



Part c: comparing shafts of various materials

```
fprintf('Part c \n');

price_lb = [0.35 1 1.8 0.4]; % price per lb of each material

weight = 2.2*[mass_AISI_1020 mass_2024_T4 mass_7075_T6
 mass_AISI_4140]; % weight of each shaft in lbs

price = price_lb.*weight; % price of each shaft

fprintf('The price of the solid AISI 1020 Steel shaft would be $%3.2f
 \n', price(1));
fprintf('The price of the solid 2024-T4 Aluminum shaft would be $%3.2f
 \n', price(2));
fprintf('The price of the solid 7075-T6 Aluminum shaft would be $%3.2f
 \n', price(3));
fprintf('The price of the solid AISI 4140 Steel shaft would be $%3.2f
 \n', price(4));

fprintf('The solid AISI 4140 Steel shaft would be the best material
 for a shaft that is light in weight and inexpensive.');
```

```
Part c
The price of the solid AISI 1020 Steel shaft would be $12.92
The price of the solid 2024-T4 Aluminum shaft would be $11.96
```

```
The price of the solid 7075-T6 Aluminum shaft would be $15.53
The price of the solid AISI 4140 Steel shaft would be $4.27
The solid AISI 4140 Steel shaft would be the best material for a shaft
 that is light in weight and inexpensive.
```

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